SET: K

PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY SOLAPUR

B.Sc.(ECS) Part- I (Semester-I) (NEP 2020) Practical Examination 2025-26

Practical Based On DSC4 – 1 Numerical Methods (G06-0104-P)

Total Marks: 30 **Time : 2 Hours**

Instructions:

1. Each question carries 10 marks.
2. Attempt **TWO** questions.
3. Use of scientific calculator is allowed.
4. Viva (05 - Marks), Certified Journal (05 - Marks)

Q.1) Prepare forward difference table and divided difference table for the data given below

X	3	5	7	9	11	13
Y=f(x)	123	354	486	695	987	1025

(10)

Q.2) Write a Python program that calculates the absolute error, relative error and percentage error between a true value and an approximate value.

(10)

Q.3) Solve the following system of linear equations by using Jacobi's iterative method.

Perform only FOUR iteration

$6X_1 + X_2 + X_3 = 20$;

$X_1 + 4X_2 - X_3 = 6$;

$X_1 - X_2 + 5X_3 = 7$

Q.4) Write a Python program to find addition and subtraction of two matrices.

(10)

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Q.1) Estimate approximate value of f (7.6) by using Newton's forward difference interpolation Formula and f (20.5) by using Newton's Backward difference interpolation formula from the following data:

X	5	10	15	20
Y=f(x)	184.7085	178.1446	171.0707	163.4328

(10)

Q.2) Write a Python program that calculates the absolute error, relative error and percentage error between a true value and an approximate value. **(10)**

Q.3) Solve the following system of linear equations by using Gauss Seidel iterative method. Perform only FOUR iterations. **(10)**

$x + 2y + 10z = 24$;
$28x + 5y - z = 32$;

$2x + 19y + 4z = 35$

Q.4) Write a Python program that performs addition , subtraction , multiplication and division of two numbers in normalized floating point forms i.e. normal forms.

(10)

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Q.1) Write a python program to create equally spaced data. Hence prepare equally spaced data for the function $f(x) = x^3 - 2x^2 + 3x + 1$ with the first argument $a = 1$ and last argument $a + nh = 5$ and interval of differencing $h = 0.5$

(10)

Q.2) Write a Python program that performs addition , subtraction , multiplication and division of two numbers in floating point forms i.e. normal forms.

(10)

Q.3) Solve the following system of linear equations by using Gauss Seidel iterative method. Perform only FOUR iterations. **(10)**

$6X_1 + X_2 + X_3 = 20$; $X_1 +$
$4X_2 - X_3 = 6$;

$X_1 - X_2 + 5X_3 = 7$

Q.4) Estimate approximate value of f (9.5) by using Newton's forward difference interpolation
formula from the following data:

(10)

X	5	10	15	20	25
Y=f(x)	22.8545	38.4563	59.1245	78.6257	89.8234